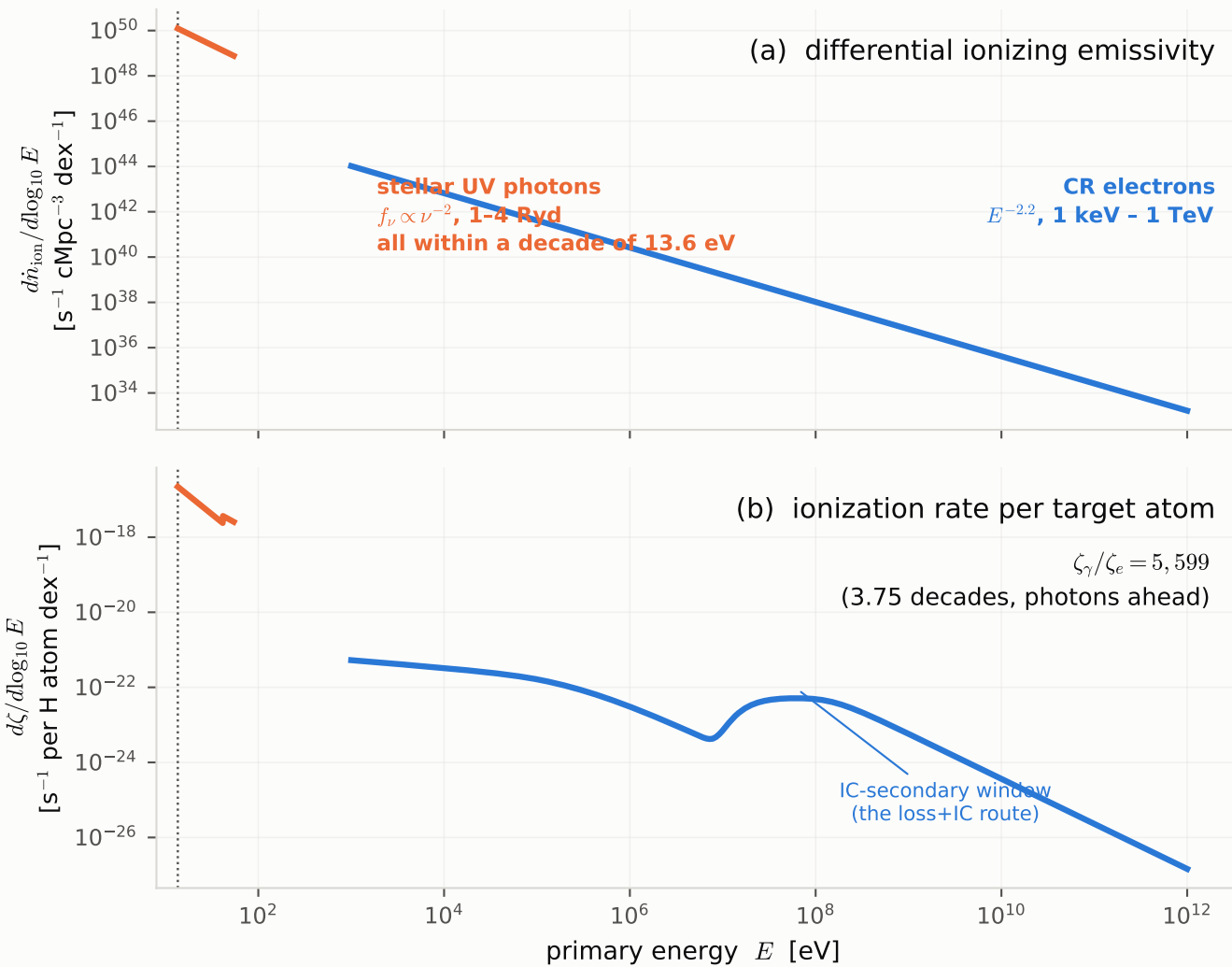


# Ionizing emissivity and ionization rate at $z = 10$ canonical parameters



IGM at  $z = 10$ ,  $x_e = 10^{-4}$ , pure H.  
Electron yields: the loss+IC route — igm\_losses.py: 7 loss mechanisms, RBED event counting, BED secondary cascade, Klein-Nishina IC, + IC-seconda  
 $\rho_{\text{UV}}$ : Donnan+24 JWST PRIMER.  $\xi_{\text{ion}} = 10^{25.28}$ : Llerena+25.  $\rho_{\text{UV}} \rightarrow \rho_{\text{SFR}}$ : Madau & Dickinson 14 (Salpeter).  
SN/ $M_\odot$  derived from the IMF, not adopted.  
Stellar SED: population of U37126 ( $z = 10.255$ , Marques-Chaves+26, BPASS v2.2.1,  $Z = 0.003$ , 6.8 Myr); LyC shape parametrised,  $\alpha$  bracketed 1-3.